BASE CHANGE FOR THE INTERNAL HOM: PULLBACK OF R-HOM IS R-HOM OF THE PULLBACKS IN THREE CASES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites seven cross-chapter labels (including a remark and a section), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Computation of R-Hom out of a strictly perfect complex. Stacks Project, Cohomology of Sheaves, Lemma lemma-Rhom-strictly-perfect (tag 0A8D). Represents the internal hom by an explicit complex of Hom sheaves.
- Pullback of finite free and strictly perfect complexes. Stacks Project, Cohomology of Sheaves, Lemma lemma-strictly-perfect-pullback (tag 09U6). Shows the comparison map is computed levelwise on free modules.

Source reference. Original remark 08I3: sources/stacks-chow/tags.txt, source lines 10801–10801. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original section 08CL: sources/stacks-chow/tags.txt, source lines 10606–10606. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08DM: sources/stacks-chow/tags.txt, source lines 10642–10642. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original section 06YI: sources/stacks-chow/tags.txt, source lines 8887–8887. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09U6: sources/stacks-chow/tags.txt, source lines 12414–12414. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GM5: sources/stacks-chow/tags.txt, source lines 20745–20745. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original section 08CA: sources/stacks-chow/tags.txt, source lines 10595–10595. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $h : X \to Y$ be a morphism of ringed spaces. Let K, M be objects of $D(\mathcal{O}_Y)$. The canonical map*

$$Lh^* R\mathcal{H}om(K, M) \longrightarrow R\mathcal{H}om(Lh^* K, Lh^* M)$$

of Remark 08I3 is an isomorphism in the following cases

1. *K is perfect,*
2. *h is flat, K is pseudo-coherent, and M is (locally) bounded below,*
3. *$\mathcal{O}_X$ has finite tor dimension over $h^{-1}\mathcal{O}_Y$, K is pseudo-coherent, and M is (locally) bounded below,*

Proof. Proof of (1). The question is local on Y, hence we may assume that K is represented by a strictly perfect complex $\mathcal{E}^\bullet$, see Section 08CL. Choose a K-flat complex $\mathcal{F}^\bullet$ representing M. Apply Lemma 08DM to see that $R\mathcal{H}om(K, L)$ is represented by the complex $\mathcal{H}^\bullet = \mathcal{H}om^\bullet(\mathcal{E}^\bullet, \mathcal{F}^\bullet)$ with terms $\mathcal{H}^n = \bigoplus_{n=p+q} \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}^{-q}, \mathcal{F}^p)$. By the construction of Lh^* in Section 06YI we see that $Lh^* K$ is represented by the strictly perfect complex $h^*\mathcal{E}^\bullet$ (Lemma 09U6). Similarly, the object $Lh^* M$ is represented by the complex $h^*\mathcal{F}^\bullet$. Finally, the object $Lh^* R\mathcal{H}om(K, M)$ is represented by $h^*\mathcal{H}^\bullet$ as $\mathcal{H}^\bullet$ is K-flat by Lemma 0GM5. Thus to finish the proof it suffices to show that $h^*\mathcal{H}^\bullet = \mathcal{H}om^\bullet(h^*\mathcal{E}^\bullet, h^*\mathcal{F}^\bullet)$. For this it suffices to note that $h^* \mathcal{H}om(\mathcal{E}, \mathcal{F}) = \mathcal{H}om(h^*\mathcal{E}, \mathcal{F})$ whenever $\mathcal{E}$ is a direct summand of a finite free $\mathcal{O}_X$-module.

Proof of (2). Since h is flat, we can compute Lh^* by simply using h^* on any complex of $\mathcal{O}_Y$-modules. In particular we have $H^i(Lh^* K) = h^* H^i(K)$ for all $i \in \mathbf{Z}$. Say $H^i(M) = 0$ for $i < a$. Let $K' \to K$ be a morphism of $D(\mathcal{O}_Y)$ which defines an isomorphism $H^i(K') \to H^i(K)$ for all $i \geq b$. Then the corresponding maps

$$R\mathcal{H}om(K, M) \to R\mathcal{H}om(K', M)$$

and

$$R\mathcal{H}om(Lh^* K, Lh^* M) \to R\mathcal{H}om(Lh^* K', Lh^* M)$$

are isomorphisms on cohomology sheaves in degrees $< a - b$ (details omitted). Thus to prove the map in the statement of the lemma induces an isomorphism on cohomology sheaves in degrees $< a - b$ it suffices to prove the result for K' in those degrees. Also, as in the proof of part (1) the question is local on Y. Thus we may assume K is represented by a strictly perfect complex, see Section 08CA. This reduces us to case (1).

Proof of (3). The proof is the same as the proof of (2) except one uses that Lh^* has bounded cohomological dimension to get the desired vanishing. We omit the details. □